

Hypothesis Testing → parametric/ non-parametric/ Bayesian

19/08/2024
Para Inf.

$$X_1, \dots, X_n \sim F_\theta : \theta \in \Theta$$

Goal is to validate some conjecture as hypothesis about,
in particular, decide b/w two opposing conjectures —
null and alternative hypothesis.
(H_0) (H_1)

Ex1: In clinical trials $\begin{cases} H_0: \text{drug not effective vs.} \\ H_1: \text{drug effective} \end{cases}$

$$\equiv H_0: \mu \geq \mu_0 \text{ vs. } H_1: \mu < \mu_0$$

↓
known

$X = \text{blood pressure after taking a drug}$

Ex2: $X = \text{Lifetime of a battery}$ $E[X] = \mu$

Customer $H_0: \mu \leq \mu_0 \text{ vs. } H_1: \mu > \mu_0$ $f_X(x) = \frac{1}{\mu} e^{-x/\mu}$

↓
(1000 hrs.)

Manufacturer $H_0: \mu \geq \mu_0 \text{ vs. } H_1: \mu < \mu_0$

General Setup:

Null Hypothesis: Any conjecture about θ , say $H_0: \theta \in \Theta_0$
where $\Theta_0 \subset \Theta$

Alternate Hypothesis: Any conjecture opposing null, say
 $H_1: \theta \in \Theta_1$ where $\Theta_1 \subset \Theta$, $\Theta_0 \cap \Theta_1 = \emptyset$

Given $\underline{X} = \underline{x}$, we want to come up with a "decision rule"

ie, reject H_0 or (accept H_0) fail to reject H_0 .

That is, we partition sample space $\mathcal{X}$ into two regions

(1) Rejection/Critical Region R

(e.g., $R = \{x: \bar{x} > c\}$)

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$$

$$H_0: \mu \leq 0$$

$$H_1: \mu > 0$$

(2) Acceptance Region $A = R^c$

$$A = \{x: \bar{x} \leq c\}$$

Intuitive Test

Reject H_0 if $\bar{x} > c$

(to find)

Our decision rule is according as a function
called test function/critical function

$$\phi(x) = \begin{cases} 1 & \text{if } x \in R \\ 0 & \text{if } x \in A \end{cases}$$

→ non-randomized test.

Given x , $\phi(x)$ is the prob. with what we reject H_0
b/c $\phi \in \{0, 1\}$

Def: Any (Borel) fn. $\phi: \mathcal{X} \rightarrow [0, 1]$ that is a statistic,
can be called a test function.

If $\phi(x) \in \{0, 1\}$ → non-randomized test
prob with what we reject H_0 ← $\phi(x) \in [0, 1]$ (and $0 < \phi(x) < 1$ for some x)

→ randomized test
We will discuss randomized test in detail later.

We do not give equal importance to H_0 and H_1 .

We assume H_0 true before the experiment. If there is enough evidence against H_0 , we reject H_0 . If we fail to reject H_0 . If no data is available, then we have no option but to "accept H_0 " or "fail to reject H_0 ".

Errors in decision rule

Decision \ True	H_0 true	H_1 true
Reject H_0	Type I	✓
Accept H_0	✓	Type II

$$X \sim f_\theta : \theta \in \Theta$$

$$H_0 : \theta \in \Theta_0 \text{ vs. } H_1 : \theta \in \Theta_1$$

$$\Theta_0 \cap \Theta_1 = \emptyset$$

$$P_\theta(\text{Type I error}) = P_\theta(\text{Reject } H_0 \text{ when } \theta \in \Theta_0)$$

$$P_\theta(\text{Type II error}) = P_\theta(\text{Accept } H_0 \text{ when } \theta \in \Theta_1)$$

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ $H_0 : \mu \leq 0$
 $H_1 : \mu > 0$

$$\phi(\bar{x}) = \begin{cases} 1 & \text{if } \bar{x} > c \\ 0 & \text{if } \bar{x} \leq c \end{cases}$$

$$P_\mu(\text{Type I error}) = P_\mu(\bar{X} \geq c) \quad \forall \mu \leq 0 \quad (\forall \mu \in \Theta_0)$$

$$P_\mu(\text{Type II error}) = P_\mu(\bar{X} \leq c) \quad \forall \mu > 0 \quad (\forall \mu \in \Theta_1)$$

Ideally: Want to have both error probabilities to be 0.

possible only in certain extreme scenarios,
but not possible in general.

Ex. $X \sim f$ Test: $H_0 : f \equiv U(0,1)$
 $H_1 : f \equiv U(2,3)$

$$\phi(x) = \begin{cases} 1 & \text{if } x > 1.5 \\ 0 & \text{if } x \leq 1.5 \end{cases} \quad \begin{aligned} P(\text{Type I Error}) &= P_{H_0}(X > 1.5) = 0 \\ P(\text{Type II Error}) &= P_{H_1}(X < 1.5) = 0 \end{aligned}$$

Minimizing both error probabilities is not possible in general
 b/c. if we reduce the rejection region R , then the acceptance
 region A is increased $\Rightarrow$

$$P(\text{Type I error}) \downarrow \text{ but } P(\text{Type II error}) \uparrow$$

Ex. $P_{\mu}(\bar{X} > c)$ $P_{\mu}(\bar{X} \leq c)$

Best we can do (By analogy with judicial system)

Let's consider Type I error to be more serious.

Strategy: First control $P(\text{Type I error}) \leq \alpha$
 The best test ϕ^* is such that it $\uparrow$
 minimizes the $P(\text{Type-II Error})$ prefixed level of
 among all tests that ensures significance
 $P(\text{Type I error}) \leq \alpha$ (eg $\alpha = 0.01, 0.05, 0.1$ etc.)

General Intuition of Constructing Test

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ $\underbrace{\quad}_{\text{known}}$

$H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$ (known)

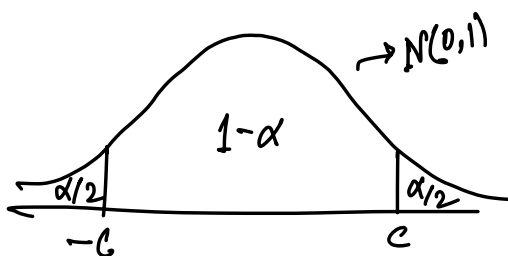
1. Start with $\bar{X}$
2. Come up with a "test statistic" that summarizes evidence against H_0 , and whose distⁿ (null distⁿ) is known when H_0 is true

$$T(\bar{X}) = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \underset{H_0}{\sim} N(0, 1)$$

Intuitive rule:

$$\phi(\underline{x}) = I\left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma_0} > c\right) \rightarrow \text{const.}$$

$$\text{Find } c : \mathbb{P}_{H_0}\left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma_0} > c\right) = \alpha$$



$\frac{c}{\sigma_0} = z_{\alpha/2}$ = upper $\alpha/2$ th quantile of $N(0,1)$

$$= (1 - \alpha/2)^{\text{th}} \text{ quantile of } N(0,1)$$

1. Test $\mu = \mu_0$ vs. $\mu > \mu_0$

$$\phi(\underline{x}) = I\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > c\right)$$

$$\mathbb{P}(\text{Type I}) = \alpha$$

Setup: $X \sim F_\theta$, $\theta \in \Theta$. $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$

Defⁿ: For any $\theta \in \Theta$, power function of a test ϕ is $\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)] \quad \forall \theta \in \Theta$

If $\phi \in \{0,1\}$ then $\beta_\phi(\theta) = \mathbb{P}_\theta(\text{rejecting } H_0) \quad \forall \theta \in \Theta$

Power of a test ϕ at θ is $\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)]$ for $\theta \in \Theta_1$

$\phi_\phi(\theta) = \mathbb{P}_\theta(\text{reject } H_0) \quad \forall \theta \in \Theta_1$
 $= \mathbb{P}(\text{rejecting a false } H_0)$

$$P_\theta(\text{Type I error}) = \beta_\phi(\theta) \text{ for } \theta \in \Theta_0$$

$$P_\theta(\text{Type II error}) = 1 - \beta_\phi(\theta) \text{ for } \theta \in \Theta_1$$

Consistent Test

A test ϕ ($\equiv \phi_n$, depends on $X_1, \dots, X_n$) is called consistent if $\beta_{\phi_n}(\theta) \rightarrow 1$ as $n \rightarrow \infty$ $\forall \theta \in \Theta_1$

As we gather more info/data, power of the test should $\rightarrow 1$

Ex: $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma_0^2)$, σ_0 known
 $H_0: \mu = \mu_0$ vs. $H_1: \mu > \mu_0$

$$\phi_n(\alpha) = I\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > z_\alpha\right)$$

$$\begin{aligned} \beta_{\phi_n}(\mu) &= P_\mu\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > z_\alpha\right) \\ &= P_\mu\left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma_0} > z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}\right) \\ &= 1 - \Phi\left(z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}\right) \end{aligned}$$

$$\xrightarrow{\quad} 1$$

= power at μ as $n \rightarrow \infty$

$$\text{Check whether } \phi_n = I\left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma_0} > z_{\alpha/2}\right)$$

is consistent for testing $\mu = \mu_0$ vs. $\mu \neq \mu_0$

Defⁿ
(1) Size of a test ϕ is $\boxed{\sup_{\theta \in \Theta_0} \beta_{\phi}(\theta)}$ = worst prob of Type I error

(2) A test ϕ is called a size- α (level- α) test if

$$\sup_{\theta \in \Theta_0} \beta_{\phi}(\theta) = \alpha \quad \left(\sup_{\theta \in \Theta_0} \beta_{\phi}(\theta) \leq \alpha \right)$$

$$\{\text{size-}\alpha \text{ tests}\} \subseteq \{\text{level-}\alpha \text{ tests}\}$$

Ex: $f_0(x) = \frac{1}{15} \quad \forall x = 0, 1, \dots, 14$

$$f_1(x) = \binom{14}{x} \left(\frac{1}{2}\right)^{14} \quad \forall x = 0, 1, \dots, 14$$

$$\phi(x) = \begin{cases} 1 & \text{if } x \leq 1 \\ 0 & \text{if } x > 1 \end{cases} \quad \alpha = 0.2$$

$$\mathbb{E}_{H_0}(\phi(X)) = \frac{2}{15} < \alpha = 0.2$$

level- α but not size- α
(cannot be attained)